INCH-POUND

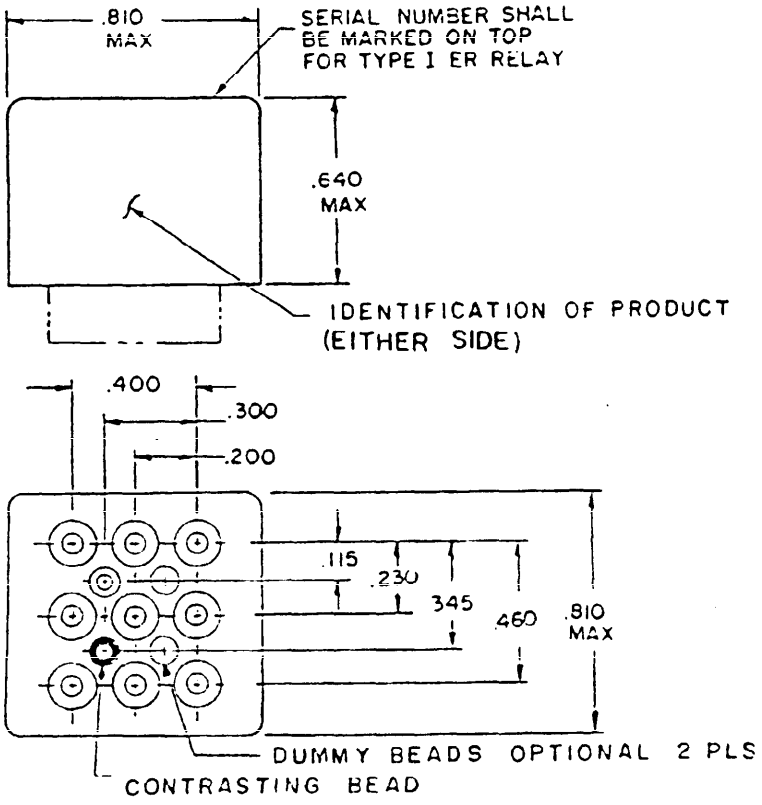
MIL-R-6106/29D
21 December 1990
SUPERSEDING
MIL-R-6106/29C
26 December 1989

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, (INCLUDING ESTABLISHED RELIABILITY (ER) TYPES), PERMANENT MAGNET DRIVE, LOW LEVEL TO 10 AMPERES, 3 PDT, ALL WELDED, HERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.



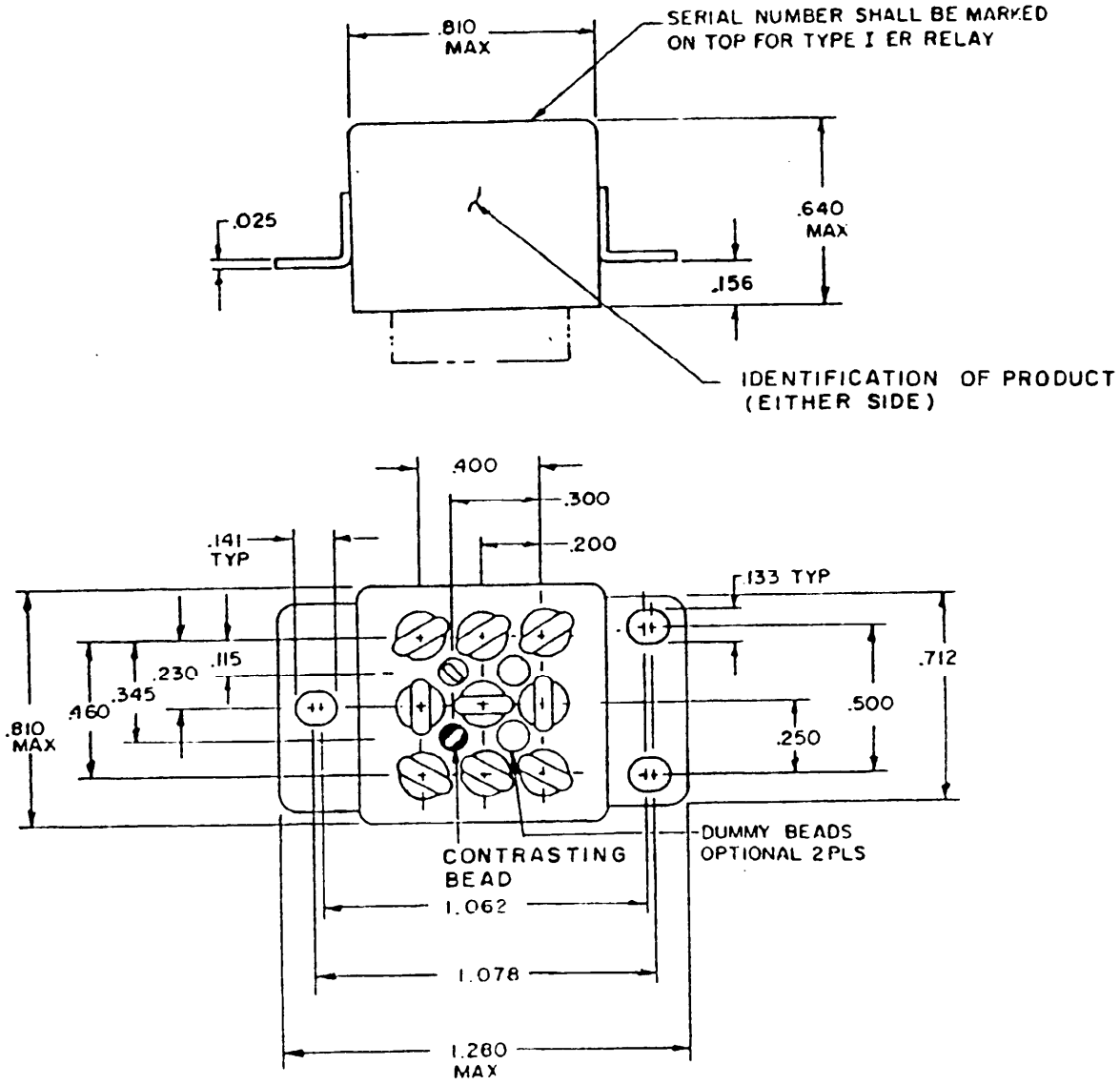
Configuration A

FIGURE 1. Outline drawing (for details, see table I).

(D) denotes change

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

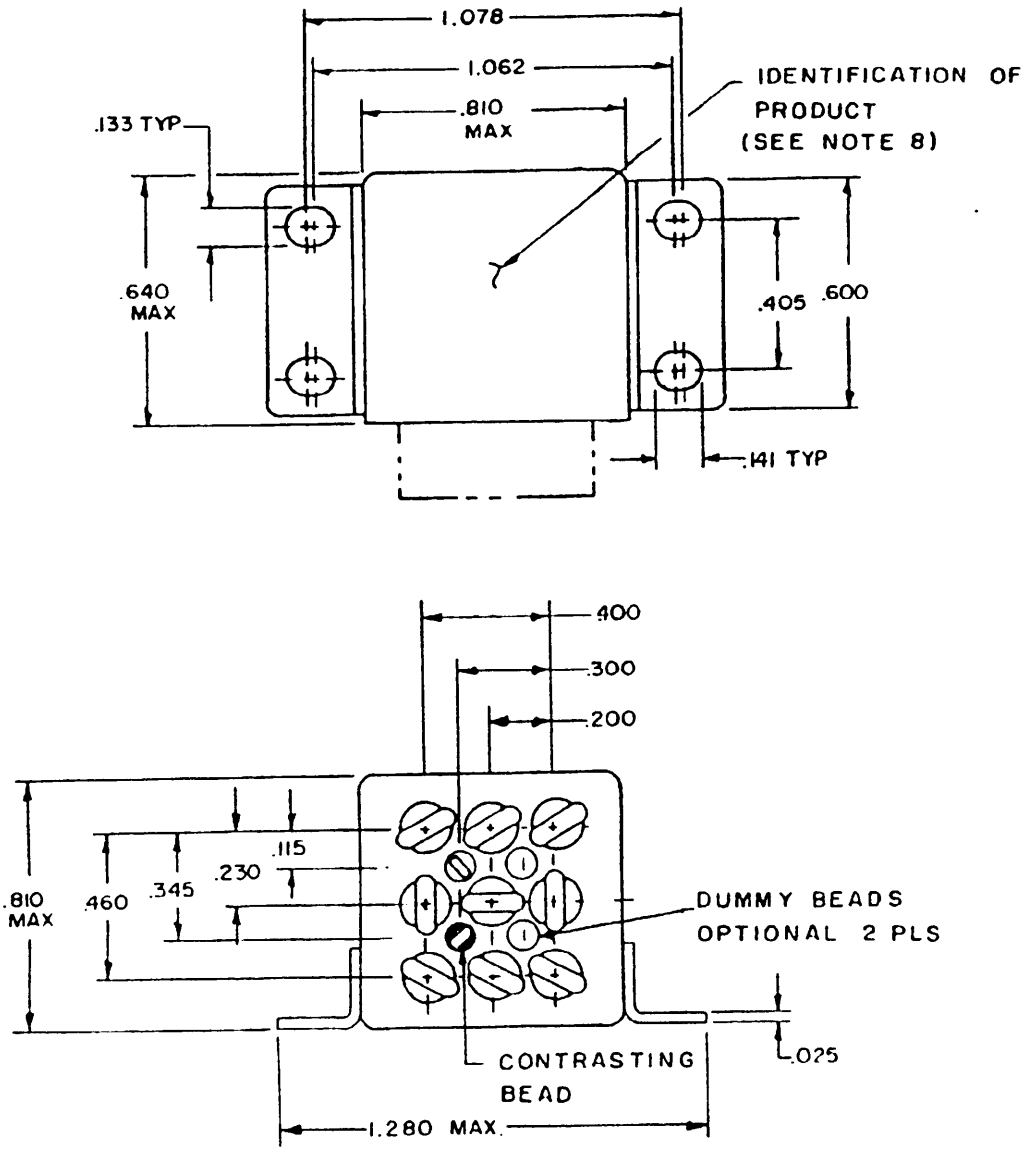
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Configuration B

FIGURE 1. Outline drawing (for details, see table I) - Continued.

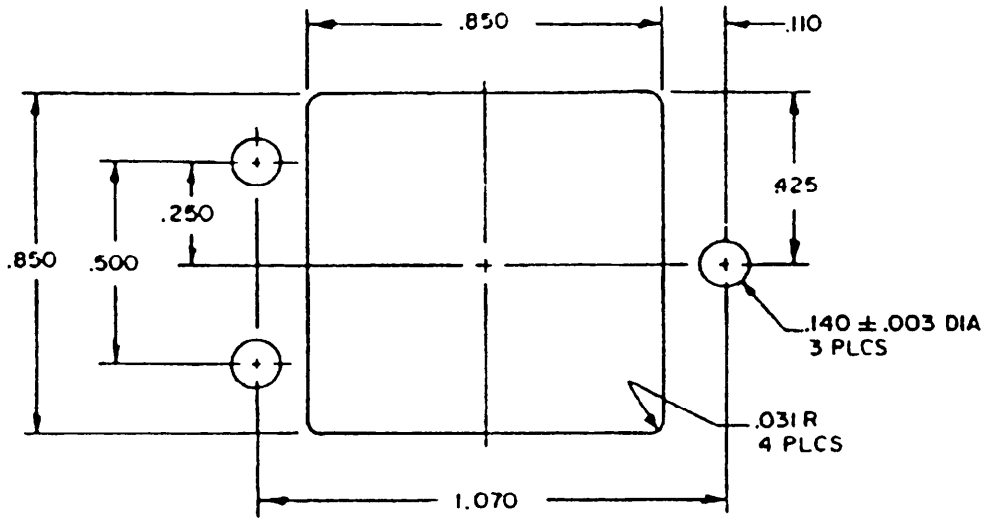
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Configuration C

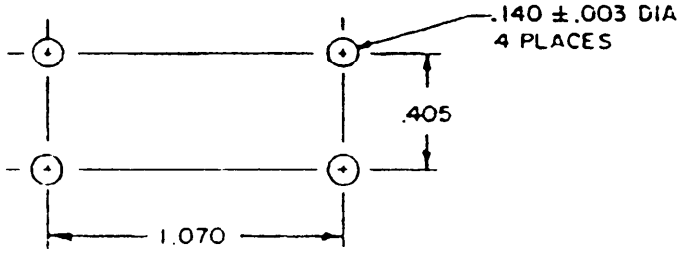
FIGURE 1. Outline drawing (for details, see table I) - Continued.

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Mounting hole layout

Configuration B



Mounting hole layout

Configuration C

FIGURE 1. Outline drawing (for details, see table I) - Continued.

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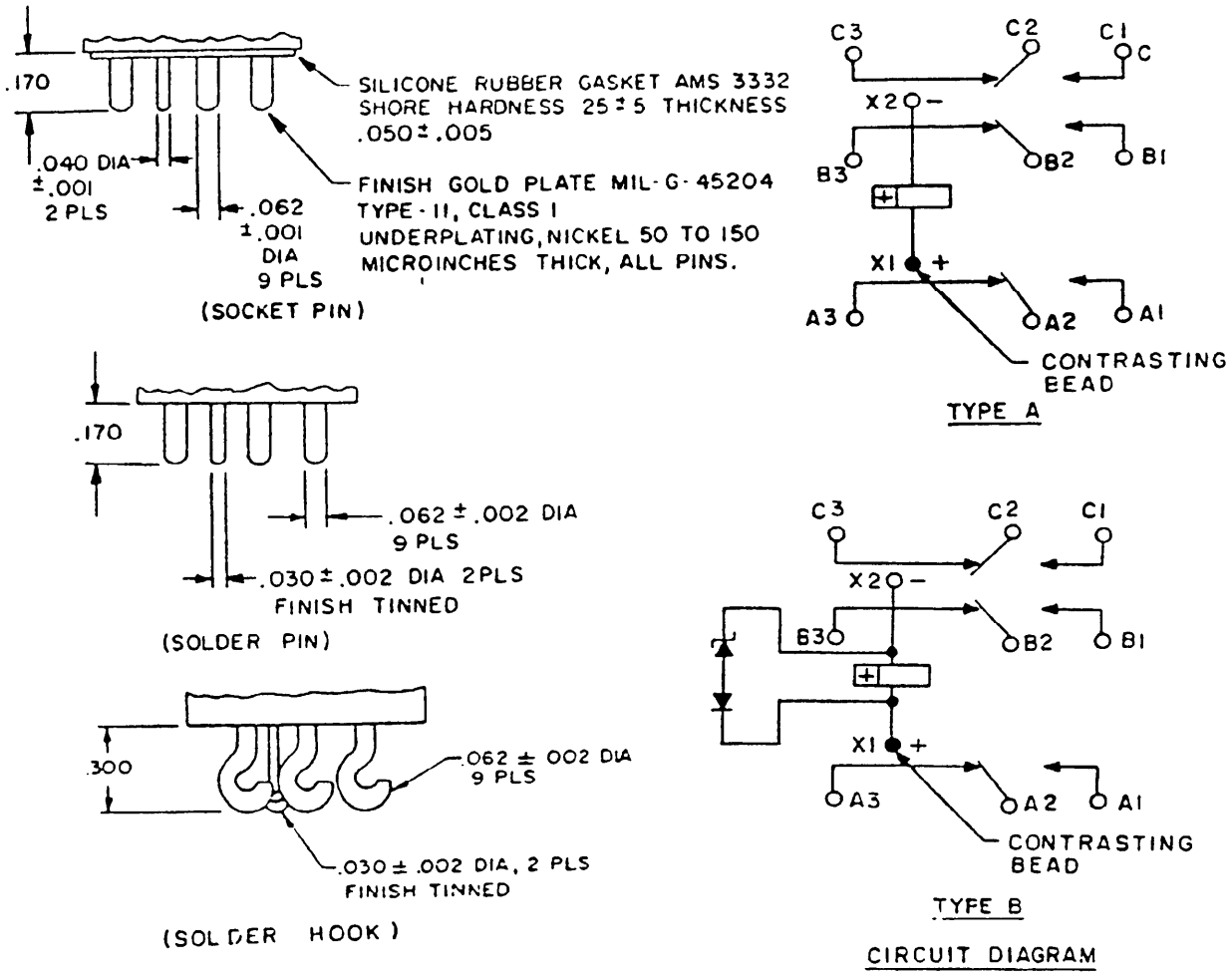


FIGURE 1. Outline drawing (for details, see table I) - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.110	2.79	.230	5.84	.600	15.24
.002	0.05	.115	2.92	.250	6.35	.640	16.26
.003	0.08	.133	3.38	.300	7.62	.712	18.08
.025	0.64	.140	3.56	.345	8.76	.810	20.57
.030	0.76	.141	3.58	.400	10.16	.850	21.59
.031	0.79	.156	3.96	.405	10.29	1.062	26.97
.040	1.02	.170	4.32	.425	10.80	1.078	27.38
.050	1.27	.200	5.08	.500	12.70	1.280	32.51

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. These relays are polarized monostable.
6. This relay shall not operate or be damaged by reverse polarity. Semiconductors shall not be used for this purpose.
7. Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electromagnetic flux.
8. Applicable to configuration C only. The circuit diagram, manufacturer's part number, and the military PIN shall be marked on the near side. The remaining portion of the nameplate data shall be marked on the far side. Relays shall be marked with the manufacturer's name or source code and data code. In addition, ER relays shall be marked with the serial number. Marking shall be with the bottom of print adjacent to the near side.

FIGURE 1. Outline drawing (for details, see table I) - Continued.

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Ⓓ TABLE I. Mechanical and physical characteristics. 1/

Military PIN 2/	Type	Coil suppression (transient voltage, back EMF) 2/	Terminal type	Mating socket	Mounting	Circuit type
M6106/29-001 3/	I	N/A	Solder pin	N/A	Configuration A	A
-002	I ER	N/A	"	"	"	A
-003 3/	I	42 V dc max	"	"	"	B
-004	I ER	42 V dc max	"	"	"	B
-005 3/	I	N/A	Solder hook	"	"	A
-006	I ER	N/A	"	"	"	A
-007 3/	I	42 V dc max	"	"	"	B
-008	I ER	42 V dc max	"	"	"	B
-009 3/	I	N/A	Socket pin		"	A
-010	I ER	N/A	"	MIL-S-12883/	"	A
-011 3/	I	42 V dc max	"	46-01	"	B
-012	I ER	42 V dc max	"		"	B
-013 3/	I	N/A	Solder pin	N/A	Configuration B	A
-014	I ER	N/A	"	"	"	A
-015 3/	I	42 V dc max	"	"	"	B
-016	I ER	42 V dc max	"	"	"	B
-017 3/	I	N/A	Solder hook	"	"	A
-018	I ER	N/A	"	"	"	A
-019 3/	I	42 V dc max	"	"	"	B
-020	I ER	42 V dc max	"	"	"	B
-021 3/	I	N/A	Socket pin		"	A
-022	I ER	N/A	"	MIL-S-12883/	"	A
-023 3/	I	42 V dc max	"	46-01	"	B
-024	I ER	42 V dc max	"		"	B
-025 3/	I	N/A	Solder pin	N/A	Configuration C	A
-026	I ER	N/A	"	"	"	A
-027 3/	I	42 V dc max	"	"	"	B
-028	I ER	42 V dc max	"	"	"	B
-029 3/	I	N/A	Solder hook	"	"	A
-030	I ER	N/A	"	"	"	A
-031 3/	I	42 V dc max	"	"	"	B
-032	I ER	42 V dc max	"	"	"	B

1/ For type I ER relays, a suffix letter, M, U, X, or P to indicate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles), M = 1%, U = 0.5%, X = 0.3% and P = 0.1%.
Example: -001M ----, 003P.

2/ JANTX or equivalent screened semiconductors shall be used in ER relays with internal coil suppression. Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode. Diodes shall have a peak inverse voltage of 600 V dc minimum.

3/ Type I relays inactive for new design after 21 Dec 1990 . See table VI for cross reference data.

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REQUIREMENTS:

Coil data:

(D) Coil data: See tables I and III.

Duty rating: Continuous.

(D) Operational data: See tables III and IV.

Time current relay characteristics: Applicable (see table II).

(D) TABLE II. Time-current relay characteristics at +25°C. 1/ 2/

1	20A	1 hour
2	30A	20 seconds
3	75A	2 seconds
4	150A	.53 second

1/ Caution. Compare with time-current characteristics of the associated circuit protective device.

2/ All relays shall withstand overload and fault currents. Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of four different current levels in the sequence as listed in table II. Separate relays shall be tested at 28 V dc and 115/200 V ac 400 Hz 3 phase. Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normally closed contacts. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

Physical:

Dimensions and configuration: See figure 1.

Weight (maximum): .062 pound, 28.15 grams.

Terminal type: See figure 1 and table I.

ENVIRONMENTAL CHARACTERISTICS:	Configurations A and B	Configuration C
Temperature rating	-70°C to +125°C	-70°C to +125°C
Maximum altitude rating	300,000 feet	300,000 feet
Shock g-level	200 g	100 g
Duration	6 ms	6 ms
Maximum duration contact opening	10 µs	10 µs
Vibration - sinusoidal:		
G-level	30 g	20 g
Frequency range	70 - 3,000 Hz	57 - 3,000 Hz
Vibration - random:		
Applicable standard	MIL-STD-202	MIL-STD-202
Method 214	0.4 g^2/Hz	0.2 g^2/Hz
Test condition	1G	1E
Duration	15 min ea plane	15 min ea plane
Acceleration	15 g	15 g

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Ⓓ TABLE III. Operating characteristics.

Dash number	Coil	Coil data										Time milliseconds (maximum)						
		Rated			Max		Max pick-up voltage			Hold voltage 2/	Drop-out voltage 2/	Operate 3/	Release 4/	Contact bounce				Break bounce NO only
														Main		Aux		
		Volts 1/	Freq. Hz	Res. Ω ±10% at 25°C	Volts	Amp	Normal 2/	High temp test	Cont current test					NO	NC	NO	NC	
-001 thru -032	X1 X2	28	DC	400	29	.081	18	19.8	22.5	7.0	1.5	6	6	1	1			.1

1/ Caution. The use of any coil voltage less than the rated voltage will compromise the operation of the relay.
2/ Over temperature range.
3/ With rated coil voltage.
4/ From rated coil voltage.

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TABLE IV. Rated contact load (amperes per pole) case grounded. 1/

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase				See appropriate footnotes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	50	10	10			10				10				
Inductive	10	6	6											
Inductive	20					8				8				
Motor	50	4	4			4				4				
Lamp	50	2	2			2				2				
Transfer load														2/
Mechanical life reduced current	200	1.25	.125			1.25				1.25				
Intermediate current	Applicable per specification													
Low level	100	Applicable per specification												3/ 4/ 5/
Multiple loads	Applicable per specification													6/

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used (make or break) above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.

2/ Transfer load indicates relay suitable for transfer between unsychronized ac power supplies at rating indicated.

3/ For low level the following shall apply:
   a. During endurance and operational reliability testing, contact load shall be 10 to 50 microamperes; 10 to 50 millivolt open circuit voltage, 100 ohm maximum contact resistance.
   b. Static contact resistance shall be performed at 50 mA maximum, 50 mV maximum.

4/ For ER relays, only the alternate low level test of operational reliability shall be used.

5/ For group A contact voltage drop test, high level testing shall be performed first, followed by low level testing. The contacts shall not make or break the high level load.

6/ Relay shall be capable of switching low level while switching any of the other rated loads on adjacent contacts.

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ELECTRICAL CHARACTERISTICS:

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric withstanding voltage (sea level):

	Initial	After life tests
Coil to case	1,000 V rms	750 V rms
Aux contacts	---	---
All other points	1,250 V rms	1,000 V rms

Dielectric withstanding voltage (altitude): 1/

	80,000 feet	300,000 feet
Coil to case	250 V rms	500 V rms
Aux contacts	---	---
All other points	250 V rms	500 V rms

Contact voltage drop or resistance:

High level:

Initial: 0.150 volt maximum.

After life: 0.175 volt maximum.

Low level:

Initial: 0.050 ohm maximum.

After life: 0.150 ohm maximum.

Overload current: 30 amperes dc, 60 amperes ac.

Rupture current: 40 amperes dc, 80 amperes ac.

Part or Identifying Number (PIN): M6106/29- (dash number from table I and a suffix letter (if applicable) indicating the failure rate level).

Supersession data: See table V.

TABLE V. Supersession data.

Superseded PIN M6106/29-	New PIN M6106/29-
-033	-025
-034	-026
-035	-027
-036	-028

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Cross-reference for Government logistical support: See table VI.

Ⓓ TABLE VI. Cross-reference for Government logistical support.

PIN M6106/29-	Support with PIN M6106/29-	PIN M6106/29-	Support with PIN M6106/29-
001	002	017	018
002	002	018	018
003	004	019	020
004	004	020	020
005	006	021	022
006	006	022	022
007	008	023	024
008	008	024	024
009	010	025	026
010	010	026	026
011	012	027	028
012	012	028	028
013	014	029	030
014	014	030	030
015	016	031	032
016	016	032	032

1/ The suffix letter M, U, X, or P to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles) M, 1.0; U, 0.5; X, 0.3; P, 0.1 (example: -002M ------- -004P).

Qualification by similarity: See table VII.

TABLE VII. Qualification by similarity. 1/

Military PIN M6106/29-	Loads		Dynamics			Environmental
	Basic qual	ER	Sockets/Pins			
	A	D	A	B	C	A
	DC 2/ 3/	DC 2/ 3/	Configuration A 2/ 3/	Configuration B 2/ 3/	Configuration C 2/ 3/	DC 2/ 4/
001 5/	4	3	2			4
002	4	3	2			4
003 5/	1, 2, 3	1, 2	1			1, 2, 3
004	2, 3	1, 2	1			2, 3
005 5/	4	3	2			4
006	4	3	2			4
007 5/	1, 2, 3	1, 2	1			1, 2, 3
008	2, 3	1, 2	1			2, 3
009 5/	4	3	2			4

See footnotes at end of table.

1/ Dielectric rating may be improved by suitable insulation of terminals and wiring after installation.

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TABLE VII. Qualification by similarity - continued. 1/

TABLE VII. Qualification by similarity - Continued. 1/

Military PIN M6106/29-	Loads		Dynamics			Environmental
	Basic qual.	ER	Sockets/Pins			
	A	D	A	B	C	A
	DC 2/ 3/	DC 2/ 3/	Configuration A 2/ 3/	Configuration B 2/ 3/	Configuration C 2/ 3/	DC 2/ 4/
010	4	3	2			4
011 5/	1, 2, 3	1, 2	1			1, 2, 3
012	2, 3	1, 2	1			2, 3
013 5/	4	3		2		4
014	4	3		2		4
015 5/	1, 2, 3	1, 2		1		1, 2, 3
016	2, 3	1, 2		1		2, 3
017 5/	4	3		2		4
018	4	3		2		4
019 5/	1, 2, 3	1, 2		1		1, 2, 3
020	2, 3	1, 2		1		2, 3
021 5/	4	3		2		4
022	4	3		2		4
023 5/	1, 2, 3	1, 2		1		1, 2, 3
024	2, 3	1, 2		1		2, 3
025 5/	4	3			2	4
026	4	3			2	4
027 5/	1, 2, 3	1, 2			1	1, 2, 3
028	2, 3	1, 2			1	2, 3
029 5/	4	3			2	4
030	4	3			2	4
031 5/	1, 2, 3	1, 2			1	1, 2, 3
032	2, 3	1, 2			1	2, 3

1/ Reference amendment 1, MIL-R-6106, for guidelines established for determining how relays are to be grouped and ranked within each subgroup.

2/ DC suppressed coils - discretes qualify only discretes; hybrids qualify only hybrids.

3/ Terminals - socket pin type terminals are considered worst cases. Solder pins and hooks may be qualified by similarity.

4/ Socket pins (with gasket seals) are considered nonsimilar to solder pins and hooks and must be tested separately.

5/ Inactive for new design after _____.

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(D) Quality conformance inspection:

Performance of group C tests not applicable to type I relays.

CONCLUDING MATERIAL

Custodians:
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-0838)

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